

VoxType

Setup Guide

v0.1.0

Hold a key. Speak. Text appears.

Offline voice-to-text powered by OpenAI Whisper
macOS • Windows • Linux

- ▶ 100% offline — no cloud, no API keys, no data leaves your machine
- ▶ GPU-accelerated on macOS (Apple Silicon MLX) and NVIDIA (CUDA)
- ▶ Hindi romanization (Devanagari → Roman script / Hinglish)
- ▶ Hold-to-talk with any configurable hotkey
- ▶ Floating recording indicator on all platforms
- ▶ System tray icon with real-time state feedback
- ▶ Clipboard paste or keystroke simulation for text insertion
- ▶ Voice Activity Detection to trim silence automatically

github.com/ryzex/voxtypes

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System Requirements

Before installing VoxType, make sure your system meets these requirements:

Component	Requirement
Operating System	macOS 12+ (Monterey or later), Windows 10/11, or Linux (X11)
Python	3.10 or newer (3.11–3.14 recommended)
RAM	4 GB minimum (8 GB recommended for larger models)
Disk Space	~500 MB for base model, up to 6 GB for large-v3
Microphone	Any working microphone (built-in or external)
GPU (optional)	Apple Silicon M1+ (macOS) or NVIDIA with CUDA (Linux/Windows)
Internet	Only for initial install and model download

i VoxType runs 100% offline after setup. No internet connection is needed for daily use. Your audio never leaves your machine.

Installation — macOS

Follow these steps to set up VoxType on macOS (Intel or Apple Silicon):

Step 1: Install Python

macOS may come with an outdated Python. Install Python 3.10+ using one of these methods:

Option A — Homebrew (recommended):

```
brew install python3
```

Option B — Official installer:

Download from <https://www.python.org/downloads/macos/> and run the .pkg installer.

Verify your Python version:

```
python3 --version # Should show 3.10 or newer
```

Step 2: Install Git (if needed)

```
# Xcode command-line tools include git:
xcode-select --install

# Or via Homebrew:
brew install git
```

Step 3: Clone & Install VoxType

```
git clone https://github.com/ryzex/voxtypes.git
cd voxtypes
pip3 install -e .
```

Step 4: Install GPU Acceleration (Apple Silicon)

If you have an M1, M2, M3, or M4 Mac, install the MLX backends for much faster transcription:

```
pip3 install -e ".[macos]"
```

i This installs `mlx-whisper` and `lightning-whisper-mlx` which use the Neural Engine and GPU on Apple Silicon. Transcription is 3–5x faster than CPU-only mode.

Step 5: Install Voice Activity Detection (recommended)

```
pip3 install -e ".[vad]"
```

VAD trims silence before transcription, reducing processing time and improving accuracy.

Step 6: Verify Installation

```
voxtypes --help
```

You should see the VoxType help text with all available options.

! Important: On macOS, always use `python3` and `pip3` (not `python/pip`) to ensure you're using the correct Python version.

Installation — Windows

Follow these steps to set up VoxType on Windows 10 or Windows 11:

Step 1: Install Python

1. Go to <https://www.python.org/downloads/windows/>
2. Download the latest Python 3.x installer (64-bit).
3. Run the installer. IMPORTANT: Check "Add python.exe to PATH" at the bottom of the first screen before clicking Install.
4. Verify installation in Command Prompt or PowerShell:

```
python --version    # Should show 3.10 or newer
pip --version
```

! If "python" is not recognized, you need to add it to your system PATH. Open Settings > System > About > Advanced system settings > Environment Variables and add Python to PATH.

Step 2: Install Git

1. Download Git from <https://git-scm.com/download/win>
2. Run the installer with default settings.
3. Verify: open a new terminal and run:

```
git --version
```

Step 3: Clone & Install VoxType

Open Command Prompt or PowerShell:

```
git clone https://github.com/ryzex/voxtypes.git
cd voxtypes
pip install -e .
```

Step 4: Install NVIDIA GPU Acceleration (optional)

If you have an NVIDIA GPU with CUDA support:

1. Install NVIDIA drivers from <https://www.nvidia.com/drivers>
2. Install CUDA Toolkit from <https://developer.nvidia.com/cuda-downloads>
3. Install PyTorch with CUDA support:

```
pip install torch --index-url https://download.pytorch.org/whl/cu121
```

i With CUDA, VoxType uses your GPU for transcription, making it significantly faster than CPU-only mode. Without an NVIDIA GPU, VoxType falls back to CPU automatically.

Step 5: Install Voice Activity Detection (recommended)

```
pip install -e ".[vad]"
```

Step 6: Verify Installation

```
voxtypes --help
```

Installation — Linux

Follow these steps to set up VoxType on Linux (Ubuntu/Debian, Fedora, Arch, etc.):

Step 1: Install System Dependencies

Ubuntu / Debian:

```
sudo apt update
sudo apt install python3 python3-pip python3-venv git
sudo apt install portaudio19-dev # Required for audio recording
sudo apt install xclip           # Required for clipboard paste
sudo apt install python3-tk      # Required for overlay indicator
```

Fedora:

```
sudo dnf install python3 python3-pip git
sudo dnf install portaudio-devel
sudo dnf install xclip
sudo dnf install python3-tkinter
```

Arch Linux:

```
sudo pacman -S python python-pip git portaudio xclip tk
```

Step 2: Clone & Install VoxType

```
git clone https://github.com/ryzex/voxtypes.git
cd voxtypes
pip3 install -e .
```

Step 3: Install NVIDIA GPU Acceleration (optional)

If you have an NVIDIA GPU:

```
# Install NVIDIA drivers (Ubuntu example)
sudo apt install nvidia-driver-535

# Install PyTorch with CUDA
pip3 install torch --index-url https://download.pytorch.org/whl/cu121
```

Step 4: Install Voice Activity Detection (recommended)

```
pip3 install -e ".[vad]"
```

Step 5: Verify Installation

```
voxtype --help
```

! VoxType requires an X11 display server for global hotkeys and clipboard. Wayland is not yet supported. If you're on Wayland, try running with XWayland: `GDK_BACKEND=x11 voxtype`

First Run & Permissions

Start VoxType from your terminal:

```
voxtype
```

On first run, the Whisper model will be downloaded automatically (~150 MB for the base model). A system tray icon will appear indicating VoxType is ready.

macOS Permissions

macOS requires explicit permission for VoxType to capture keyboard input, record audio, and type text. Go to System Settings > Privacy & Security and enable:

- Accessibility — required for the global hotkey listener and text insertion via clipboard paste
- Input Monitoring — required for detecting hotkey press/release events
- Microphone — required for audio recording

How to grant permissions:

1. Open System Settings > Privacy & Security
2. Click Accessibility, then click the + button
3. Add your terminal app (Terminal, iTerm2, VS Code, etc.)
4. Repeat for Input Monitoring and Microphone

- i You may need to restart VoxType (and sometimes your terminal) after granting permissions for them to take effect.

Windows Permissions

Windows generally requires no special setup. However:

- If hotkeys don't work in elevated (admin) applications, run your terminal as Administrator
- Windows may ask for microphone access on first use — click Allow
- Windows Defender may warn about an unknown app — click More info > Run anyway

Linux Permissions

On Linux, ensure:

- X11 is your display server (check with: `echo $XDG_SESSION_TYPE`)
- xclip or xsel is installed for clipboard text insertion
- Your user has microphone access (usually automatic; check with: `arecord -l`)
- If using PipeWire, ensure PulseAudio compatibility layer is enabled

Configuration

VoxType stores its configuration at:

```
~/.voxtypes/config.json          # macOS / Linux
%USERPROFILE%\voxtypes\config.json # Windows
```

You can edit this file directly or use CLI flags (which override the config file). Here are all available options:

Option	Default	Description
hotkey	right_ctrl	Key to hold for recording
whisper_model	base	Whisper model size (tiny/base/small/medium/large-v3)
whisper_device	auto	Compute device (auto/cpu/cuda)
whisper_compute_type	int8	Model precision (int8/float16/float32)
quantization	null	MLX quantization (null/4bit/8bit) — macOS only
language	null	Force language (null = auto-detect)
typing_method	clipboard	Text insertion method (clipboard/simulate)
audio_sample_rate	16000	Audio sample rate in Hz

Option	Default	Description
audio_channels	1	Number of audio channels
vad_filter	true	Enable voice activity detection
beam_size	5	Beam search width for decoding
overlay_enabled	true	Show floating recording indicator
romanize	false	Convert Hindi Devanagari to Roman script
initial_prompt	null	Prompt to guide Whisper transcription style

Example config file:

```
{
  "hotkey": "right_shift",
  "whisper_model": "small",
  "language": "en",
  "typing_method": "clipboard",
  "overlay_enabled": true,
  "vad_filter": true
}
```

Whisper Models

Whisper comes in several sizes. Larger models are more accurate but slower and use more memory:

Model	Parameters	VRAM	Speed	Best For
tiny	39M	~1 GB	Fastest	Quick tests, low-resource devices
base	74M	~1 GB	Fast	Daily use, good accuracy (recommended)
small	244M	~2 GB	Moderate	Better accuracy, still fast on GPU
medium	769M	~5 GB	Slower	High accuracy, multilingual
large-v3	1550M	~10 GB	Slowest	Best accuracy, all languages

i For most users, the 'base' model offers the best balance of speed and accuracy. Use 'small' if you need better quality and have a GPU.

Hindi Romanization

VoxType includes built-in Hindi romanization (Hinglish mode). When enabled, Hindi speech is transcribed in

Devanagari by Whisper, then automatically converted to Roman script.

How It Works

1. Whisper transcribes Hindi speech to Devanagari: "मैं घर जा रहा हूँ"
2. The romanizer converts it to: "main ghar ja raha hoon"
3. A 1,800+ word Hinglish dictionary ensures natural transliteration for common words.

Enabling Romanization

Set romanize to true in your config file:

```
{
  "romanize": true
}
```

Or use the CLI flag:

```
voxtypes --romanize
```

i When romanize is on and Whisper detects Urdu instead of Hindi, VoxType automatically re-transcribes with Hindi forced, since Devanagari has explicit vowels needed for accurate romanization.

Whisper Backends & Speed Tiers

VoxType supports multiple Whisper backends. The app automatically selects the fastest available backend for your platform:

Backend	Platform	Device	Speed	Notes
lightning-whisper-mlx	macOS (Apple Silicon)	GPU (MLX)	Fastest	4-bit/8-bit quantization
mlx-whisper	macOS (Apple Silicon)	GPU (MLX)	Fast	Full-precision MLX
faster-whisper (CUDA)	Linux / Windows	NVIDIA GPU	Fast	CTranslate2 + CUDA
faster-whisper (CPU)	All platforms	CPU	Moderate	Universal fallback

- i On macOS with Apple Silicon, the app automatically uses MLX backends for GPU acceleration. No configuration needed — just install the macos extras with: `pip3 install -e ".[macos]"`

Disk Space Requirements

Here's how much disk space each component requires:

Component	Size	Location
VoxType app	~5 MB	Installation directory
Python dependencies	~200–500 MB	Site-packages
Whisper tiny model	~75 MB	~/.cache/huggingface/hub/
Whisper base model	~150 MB	~/.cache/huggingface/hub/
Whisper small model	~500 MB	~/.cache/huggingface/hub/
Whisper medium model	~1.5 GB	~/.cache/huggingface/hub/
Whisper large-v3 model	~3 GB	~/.cache/huggingface/hub/
MLX backends (macOS)	~200–400 MB	Site-packages
Silero VAD model	~5 MB	Torch hub cache

CLI Commands & Options

VoxType is launched from the command line. All config file options can be overridden with CLI flags:

```
voxtypes # Run with defaults
voxtypes --model small # Use small model
voxtypes --hotkey right_shift # Use right Shift as hotkey
voxtypes --language en # Force English
voxtypes --typing-method simulate # Type via key simulation
voxtypes --romanize # Enable Hinglish mode
voxtypes --quantization 4bit # MLX 4-bit quantization
voxtypes -v # Verbose logging
voxtypes -vv # Debug logging
```

Flag	Config Key	Description
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Flag	Config Key	Description
--model	whisper_model	Whisper model size
--device	whisper_device	Compute device (auto/cpu/cuda)
--compute-type	whisper_compute_type	Model precision
--quantization	quantization	MLX quantization level
--hotkey	hotkey	Hold-to-talk key name
--language	language	Force transcription language
--typing-method	typing_method	Text insertion method
--romanize	romanize	Enable Hindi romanization
--beam-size	beam_size	Beam search width
--no-overlay	overlay_enabled	Disable floating indicator
-v / -vv	—	Verbose / debug logging

Troubleshooting

Hotkey not working

macOS:

- Grant Accessibility and Input Monitoring in System Settings > Privacy & Security
- Restart your terminal after granting permissions
- Try a different hotkey: `voxtype --hotkey right_shift`

Windows:

- Run your terminal as Administrator
- Some antivirus software blocks keyboard hooks — add VoxType as an exception

Linux:

- Ensure X11 is running (`echo $XDG_SESSION_TYPE` should show x11)
- Try running with `sudo` if permissions are insufficient

No audio recorded

- Check that your microphone is connected and set as the default input device
- macOS: Grant Microphone permission in System Settings > Privacy & Security
- Windows: Check Settings > Privacy > Microphone > Allow apps to access your microphone
- Linux: Verify with `arecord -l` or `pactl list sources short`

- Run with `-vv` flag to see detailed audio device information

Transcription is inaccurate

- Try a larger model: `voxtype --model small` or `voxtype --model medium`
- Ensure you're speaking clearly and close to the microphone
- Set the language explicitly: `voxtype --language en`
- Use initial prompt for domain terms: add `"initial_prompt"` to `config.json`
- Enable VAD to trim silence: `pip3 install -e ".[vad]"`

Text not appearing in target app

- Try switching typing method: `voxtype --typing-method simulate`
- Some apps block clipboard paste — keystroke simulation may work better
- Linux: Ensure `xclip` or `xsel` is installed (`sudo apt install xclip`)
- macOS: Ensure Accessibility permission is granted

High memory usage

- Use a smaller model (tiny or base)
- On macOS, MLX backends are more memory-efficient than CPU-only mode
- Close other memory-intensive applications

"Command not found" after installation

- Ensure `pip` installed to a directory in your `PATH`
- Try: `python3 -m voice_typing` (or `python -m voice_typing` on Windows)
- On macOS/Linux, you may need: `export PATH="$HOME/.local/bin:$PATH"`
- On Windows, restart your terminal after installation

Hotkey Reference

VoxType supports a wide range of hotkeys. Set your preferred key with `--hotkey` or the config file:

Key Name	Key	macOS	Windows	Linux
<code>right_ctrl</code>	Right Control	✓	✓	✓
<code>right_shift</code>	Right Shift	✓	✓	✓
<code>right_alt</code>	Right Alt / Option	✓	✓	✓
<code>left_ctrl</code>	Left Control	✓	✓	✓
<code>left_shift</code>	Left Shift	✓	✓	✓

Key Name	Key		macOS	Windows	Linux
left_alt	Left Alt / Option		✓	✓	✓
caps_lock	Caps Lock		✓	✓	✓
f13 – f20	Function keys 13–20		✓	✓	✓
scroll_lock	Scroll Lock		✗	✓	✓
pause	Pause / Break		✗	✓	✓
insert	Insert		✗	✓	✓
print_screen	Print Screen		✗	✓	✓

i On macOS, some keys (Insert, Pause, Scroll Lock, Print Screen) are not available. Use Right Ctrl, Right Shift, Right Alt, or function keys instead.